Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. We trained each model for 50 epochs with a batch size of 32.

2 Results

Figure 1 shows the validation accuracy during training.

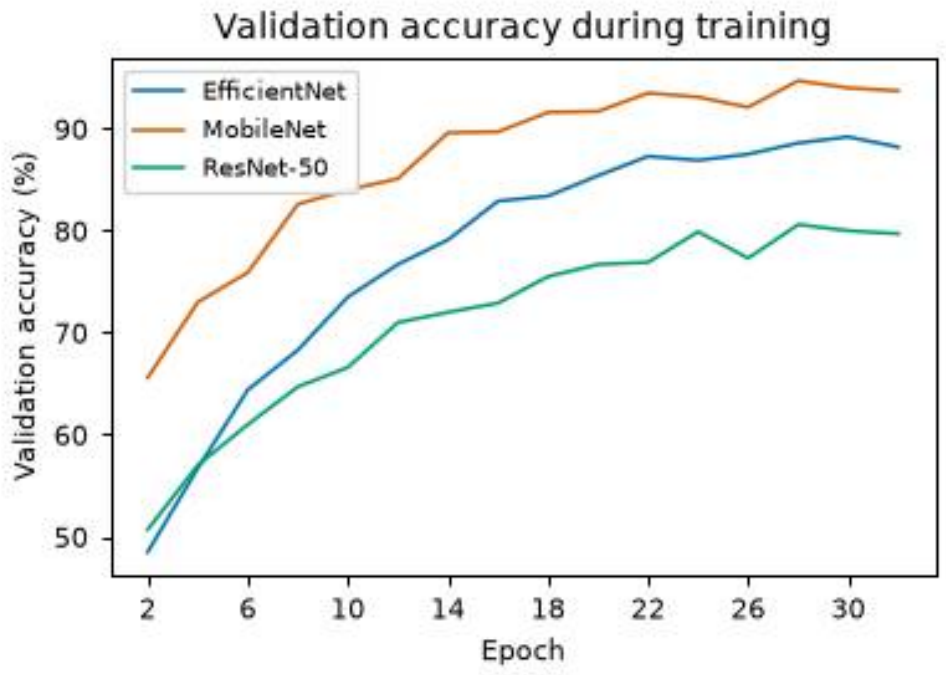


Figure 1: Validation accuracy during training for the models in this experiment.

From epoch 16 to epoch 20, the validation accuracy of MobileNet improves by a relative 7.9%. As shown in Figure 1, MobileNet outperforms EfficientNet by 12 percentage points at epoch 6.

Figure 2 shows the training loss.

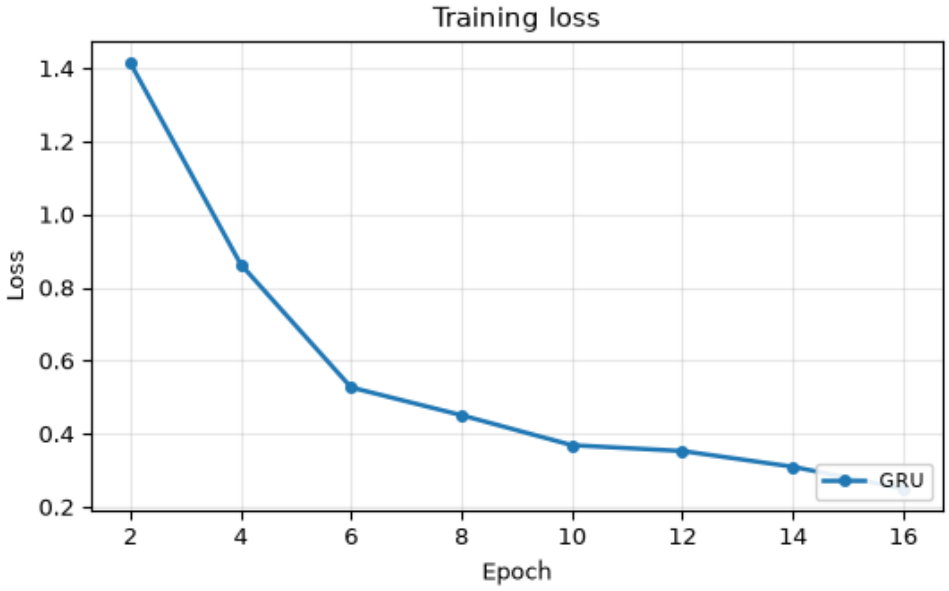


Figure 2: Training loss for the models in this experiment.

At epoch 12, the loss of GRU is 0.994. At epoch 8, the loss of GRU is 0.387. At epoch 16, the loss of GRU is

0.506.

Figure 3 shows the test accuracy of GRU by dataset.

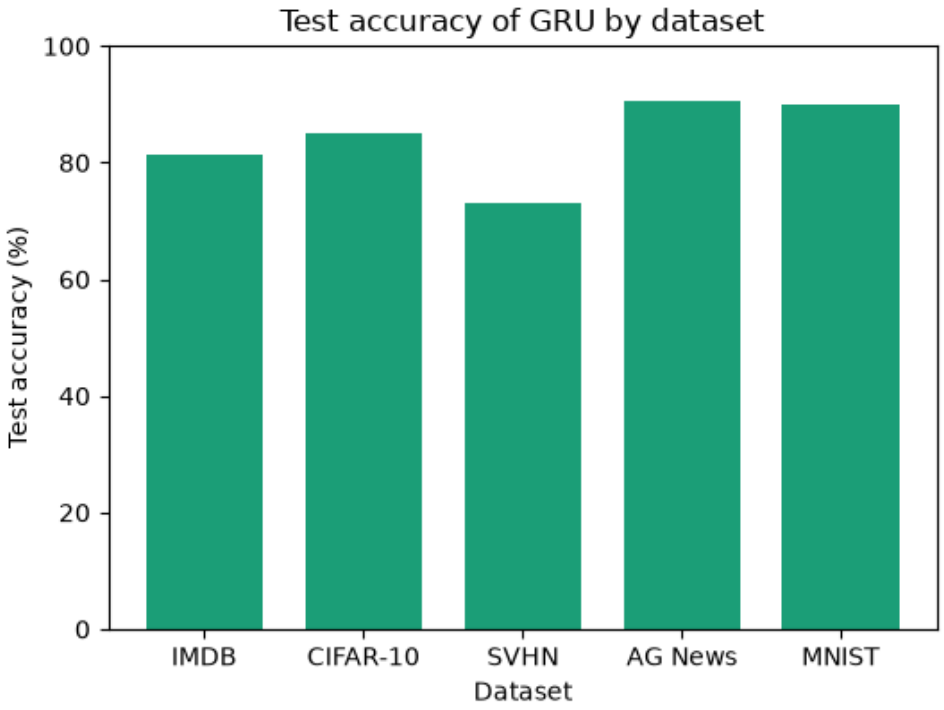


Figure 3: Test accuracy of GRU by dataset for the models in this experiment.

On AG News, GRU obtains a test accuracy of 100.6%. On CIFAR-10, GRU obtains a test accuracy of 80.2% (Figure 3). GRU reaches 92.6% test accuracy on AG News (Figure 3).

3 Discussion

On Tiny-ImageNet, Naive Bayes obtains an accuracy of 85.6%. Overall, the results are in line with what we expected from the literature.